

TESSERA: Temporal Embeddings of Surface Spectra for Earth Representation and Analysis

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Abstract

Satellite Earth-observation (EO) time series in the optical and microwave ranges of the electromagnetic spectrum are often irregular due to orbital patterns and cloud obstruction. Compositing addresses these issues but loses information with respect to vegetation phenology, which is critical for many downstream tasks. Instead, we present TESSERA, a pixel-wise foundation model for multi-modal (Sentinel-1/2) EO time series that learns robust, label-efficient embeddings. During model training, TESSERA uses Barlow Twins and sparse random temporal sampling to enforce invariance to the selection of valid observations. We employ two key regularizers: global shuffling to decorrelate spatial neighborhoods and mix-based regulation to improve invariance under extreme sparsity. We find that for diverse classification, segmentation, and regression tasks, TESSERA embeddings deliver state-of-the-art accuracy with high label efficiency, often requiring only a small task head and minimal computation. To democratize access, adhere to FAIR principles, and simplify use, we release global, annual, 10m, pixel-wise int8 embeddings together with open weights/code and lightweight adaptation heads, thus providing practical tooling for large-scale retrieval and inference at planetary scale. All code and data are available at <https://github.com/ucam-eo/tessera>.

1. Introduction

Earth Observation (EO) satellites continuously observe our planet, yet cloud cover, irregular revisit intervals depending on latitude, sensors that differ in repeat period, spatial and spectral resolutions, and noise complicate the resulting time series. For many EO applications, such as agriculture, forestry, and environmental monitoring, clouds frequently obscure the evolution of target phenomena. Al-

though compositing can yield cloud-free mosaics and reduce noise in microwave measurements, it flattens phenological dynamics and transient events that downstream models require [117].

Recent remote-sensing foundation models (RSFMs) have made substantial progress in large-scale pretraining, yet most assume heavily preprocessed cloud-filtered inputs. These predominantly patch-based models train on composites or temporal averages, a simplification that discards partly-clouded observations that contain valuable temporal information. Consequently, learned embeddings often bias toward idealized, cloud-free conditions, limiting their robustness to the irregular, incomplete sampling that characterizes operational EO time series [110].

We approach learning EO-based embeddings differently: rather than filtering imperfect observations, we use Barlow Twins [53, 116] to create embeddings that are robust to temporal sampling variability. Specifically, by forcing embeddings to be invariant to randomly selected subsets of cloud-free observations from the same location, our embeddings model consistent physical processes, which results in their generalization across sensors, seasons, and regions.

Building on work by Lisaius *et al.* [53], which describes the use of Barlow Twins for EO for optical data streams, we introduce TESSERA, a pixel-wise RSFM for multi-modal (Sentinel-1/2) sequences. Our work differs from that of Lisaius *et al.* [53] in four additional ways. First, their work created embeddings from Sentinel 2 alone, whereas TESSERA combines observations from Sentinel 1 and 2 in a novel architecture. Second, we introduce two complementary regularizers: global shuffling to disrupt short-range spatial correlations, while mix-based regularization blends observations across time and samples to stabilize learning under severe sparsity.

Third, unlike nearly all prior work, open-source global, annual, pixel-wise int8 embeddings for each 10m $\times$ 10m location on land, along with open-source model weights,

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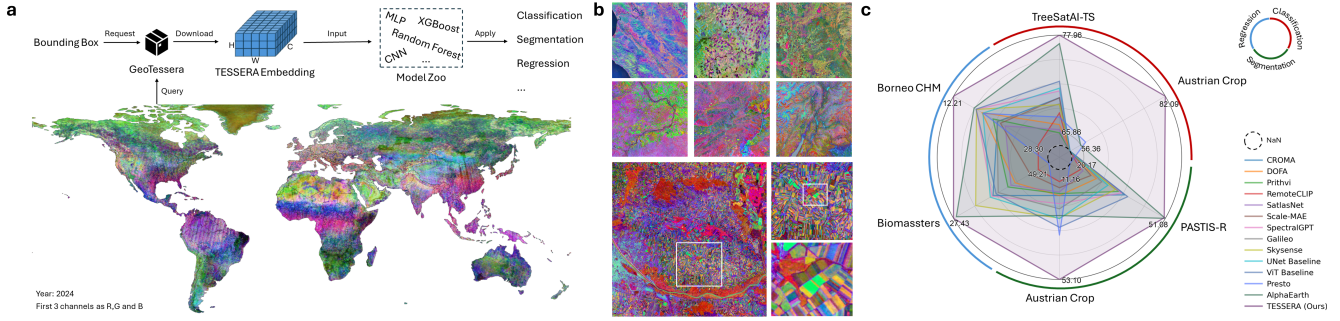


Figure 1. **TESSERA advances the embedding-as-data approach for Earth Observation.** It delivers (a) analysis-ready global-scale products, (b) pixel-wise, high-fidelity representations, and has (c) state-of-the-art downstream accuracy.

code, and the GeoTessera Python library [58] to efficiently retrieve embeddings from our data store for large-scale inference. TESSERA embeddings can be paired with lightweight adaptation heads (small MLPs or shallow convolutional models) to achieve high performance with minimal computation, enabling rapid prototyping and straightforward deployment. TESSERA thus emphasizes usability through the FAIR principles (Findable, Accessible, Interoperable, and Reusable).

Finally, we evaluate TESSERA on a wide range of classification, segmentation, and regression benchmarks spanning multiple regions and label regimes. We find that our embeddings remain consistent when calendars shift, seasons are partially missing, or cloud cover is dense. Moreover, we achieve state-of-the-art accuracy with strong label efficiency, particularly under low-label conditions, and the TESSERA embeddings degrade gracefully under heavy cloud cover. Ablations confirm that temporal sampling invariance, global shuffling, and mix-based regulation contribute each to robustness and generalization.

Our contributions are:

1. **Global, pixel-wise, label-efficient embeddings.** We use a novel self-supervised architecture to train a multi-modal (S1/S2) pixel-wise RSFM.
2. **FAIR embeddings-as-data.** We are releasing global, annual, 10 m, pixel-wise int8 embeddings with open-source weights, code, and a Python library—advancing FAIR deployment-ready resources.
3. **Comprehensive evaluation.** We compare our embeddings with a large number of other state-of-the-art foundation models and find that, for diverse classification, segmentation, and regression tasks, TESSERA embeddings deliver outstanding performance.

2. Related Work

2.1. Foundation Models for Earth Observation

Remote-sensing foundation models (RSFMs) have rapidly advanced along several methodological lines. A large body of work has used self-supervised pretraining using either

contrastive objectives (e.g., SeCo [61], SkySense [33]) or masked image modeling (e.g., SatMAE [17], ScaleMAE [73], Cross-Scale MAE [86], S2MAE [49], SatMAE++ [67]). Other variants leverage global-local view alignment or distilled objectives (e.g., DINO-MM/DINO-MC [101, 108]). Model scopes have expanded from single-modality encoders toward broader families such as RingMo and RingMo-Lite [83, 103], Satlas [8], Prithvi and Prithvi-EO-2.0 [78, 84], GFM/msGFM [34, 63], Skysense series [33, 57, 109, 122, 124]. Work on multi-sensor fusion explicitly integrates optical and SAR (e.g., CROMA [31], SkySense [33], RingMo-Sense [115]). In parallel, early efforts like MOSAIKS [74] explored lightweight universal features to lower adoption barriers.

Despite these advances, most RSFMs are trained on patch-based input and implicitly assume that the data has been preprocessed (e.g., aggressive cloud filtering or temporal averaging). Such assumptions simplify pretraining, but discard partially cloudy images that still carry useful temporal signals, biasing representations toward idealized, cloud-free conditions. Moreover, strong performance typically requires fine-tuning the backbone for each downstream task, which can be compute- and label-intensive for EO users.

2.2. Embedding-Based Foundation Models

A complementary line of work delivers *embeddings-as-data*: precomputed, generic, task-agnostic features that can be easily adapted with lightweight heads (shown in Figure 2 c). Representative models include Presto [91], which learns pixel-level temporal representations, and Google’s AlphaEarth [13], which provides precomputed embeddings as a ready-for-analysis product. This paradigm reduces computational costs and democratizes access to high performance; however, existing approaches are often not fully open-sourced, lack pixel-level granularity at a global scale, or do not explicitly account for irregular temporal sampling.

Our work follows the embedding-based paradigm while addressing these gaps. TESSERA introduces temporal sampling invariance to handle cloud- and orbital pattern induced irregularity and combines redundancy reduction with global

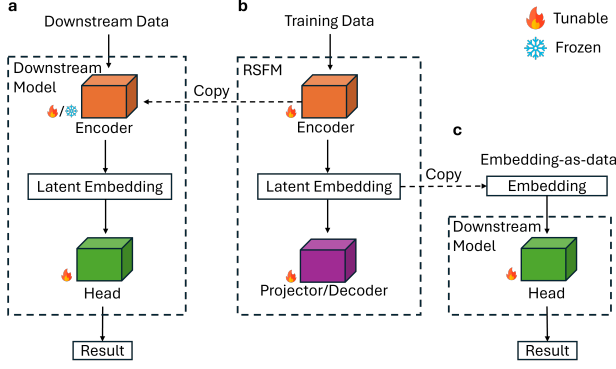


Figure 2. **Downstream adaptation paradigms.** (a–b) Most RSFMs require fine-tuning the encoder or training task-specific decoders and often do not expose intermediate embeddings. (b–c) TESSERA produces fixed, task-agnostic embeddings that plug into lightweight heads, avoiding backbone fine-tuning.

shuffling and mix-based regulation for robustness. In contrast to prior embedding releases, we provide FAIR global, annual, 10 m pixel-wise embeddings, together with open weights, code, and the GeoTessera library [58] enabling analysis-ready use without backbone fine-tuning.

3. TESSERA

3.1. d-pixel: A Representation for Temporal Data

We consider remote sensing data [27, 28] with C channels (spectral bands or polarizations). Each data tile R_t at time t is represented as a 3D array with dimensions:

$$R_t \in \mathbb{R}^{W \times H \times C}$$

where W is the width (longitude dimension), H is the height (latitude dimension), and C is the number of spectral channels.

Each tile is accompanied by a corresponding binary mask V_t of dimensions:

$$V_t \in \{0, 1\}^{W \times H}$$

where $V_t(i, j) = 0$ indicates clouding or missing data for the pixel at spatial coordinates (i, j) , and $V_t(i, j) = 1$ indicates valid data.

Temporal Data Stacking We stack resampled and spatially-aligned tiles over a time period that spans T time steps ($t = 0, 1, \dots, T-1$). The temporal data stack is defined as

$$\mathbf{D} = [R_0, R_1, \dots, R_{T-1}]$$

$$\mathbf{M} = [V_0, V_1, \dots, V_{T-1}]$$

Time Series Extraction and d-pixel Definition For a given spatial location (i, j) and spectral channel c , the time series $S_{i,j,c}$ represents all channel c values at coordinates

(i, j) over the entire time period:

$$S_{i,j,c} = [R_0(i, j, c), R_1(i, j, c), \dots, R_{T-1}(i, j, c)]$$

We define a *d-pixel* $P_{i,j}$ as the collection of all spectral channels by timesteps at a given spatial location (i, j) :

$$P_{i,j}(c) = S(i, j, c)$$

In other words, the d-pixel provides all spectral values (Sentinel-2) or backscatter values (Sentinel-1) at a given point over time. Note that d-pixels are usually sparse and are accompanied by a mask vector $m_{i,j}$ of size T that indicates the timesteps for which there are valid data, with a value of 1 indicating that the corresponding row in $P_{i,j}$ is valid. The d-pixel formulation preserves the full temporal phenology while gracefully accommodating irregular sampling.

3.2. Model Architecture

TESSERA employs a simple yet efficient dual-branch encoder architecture to process optical and radar modalities separately before fusion.

Modality Encoders. Each encoder receives a masked time series $\{(P_{i,j}^{(t)}, m_{i,j}^{(t)})\}_{t=1}^T$. Valid observations are embedded via a linear projection $\phi: \mathbb{R}^C \rightarrow \mathbb{R}^d$, and temporal context is injected via learnable Day-of-Year (DoY) positional encodings:

$$e_t = \phi(P_{i,j}^{(t)}) + \psi(\text{DoY}(t)),$$

where $\psi(\cdot)$ maps the normalized day of the year to a d -dimensional embedding space. The sequence $\{e_t\}_{t:m_{i,j}^{(t)}=1}$ is padded to a fixed length $L = 40$ via sampling with replacement if necessary, then processed by a 4-layer Transformer encoder with multi-head self-attention:

$$H = \text{TransformerEncoder}([e_{t_1}, \dots, e_{t_L}]).$$

A GRU-based pooling layer aggregates the temporal sequence into a fixed-size vector:

$$z_{\text{mod}} = \text{GRU}(H) \in \mathbb{R}^{128}.$$

Feature Fusion and Quantization. The optical embeddings (z_{S2}) and radar embeddings (z_{S1}) are concatenated and passed through a 2-layer MLP to produce the fused 128-dimensional embedding:

$$z = \text{MLP}([z_{S2}; z_{S1}]) \in \mathbb{R}^{128}.$$

We apply Quantization-Aware Training (QAT) to compress z into 8-bit integers, reducing storage by $\sim 4\times$ with negligible performance loss.

Projector Network. For training, z is expanded to 16,384 dimensions using a deep projector MLP:

$$\hat{z} = \text{Projector}(z) \in \mathbb{R}^{16384},$$

which is discarded during inference.

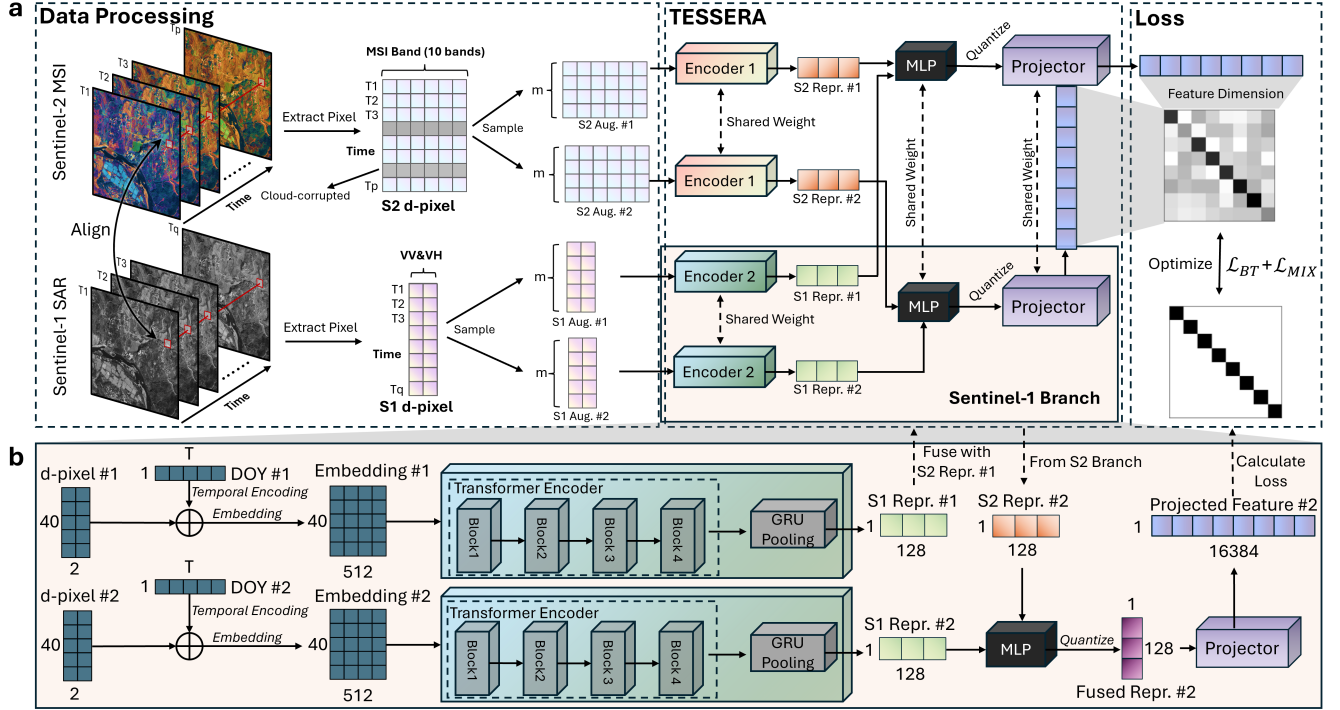


Figure 3. **Overview of the TESSERA processing pipeline.** (a) Overall architecture: multi-temporal Sentinel-1/2 observations are converted into modality-specific *d-pixels*, augmented twice, and encoded by dual branches with shared weights to produce compact 128-D embeddings. (b) Zoom-in on the Sentinel-1 branch: temporal sequences are processed by a 4-block Transformer followed by GRU pooling to capture dynamic backscatter patterns before fusion into the joint TESSERA embedding.

3.3. Pretraining

TESSERA is pretrained on ~ 800 million *d-pixels* sampled from 3,012 global MGRS tiles (2017–2024), using 16 AMD MI300X GPUs (192GB GPU RAM each). For each *d-pixel*, two augmented views (Y_A, Y_B) are generated by independently sampling 40 valid time steps from the Sentinel-1 and Sentinel-2 time series. The model is trained for one epoch with a global batch size of 32,768 using AdamW ($\eta = 0.002$, weight decay 10^{-6}), with a linear warmup followed by cosine decay.

The network processes these two views (Y_A, Y_B) through the dual-encoder and the projector to produce batch-normalized embeddings Z_A and Z_B . The standard Barlow Twins loss function, $\mathcal{L}_{BT}$, is defined as [116]:

$$\mathcal{L}_{BT} = \sum_i (1 - C_{ii})^2 + \lambda_{BT} \sum_i \sum_{j \neq i} C_{ij}^2$$

Here, C is the cross-correlation matrix computed between the batch-normalized embeddings Z_A and Z_B . The indices i and j iterate over the dimensions of the embedding vectors. The first term (invariance term) encourages similar embeddings for different views of the same input ($C_{ii} \rightarrow 1$). The second term (redundancy reduction term) promotes information efficiency by minimizing the correlation between different embedding dimensions ($C_{ij} \rightarrow 0$ for $i \neq j$),

weighted by λ_{BT} . We used the value of $\lambda_{BT} = 5 \times 10^{-3}$ as recommended in Reference [116].

To further enhance the robustness of the model and mitigate overfitting, TESSERA incorporates an additional mix-up regularization term $\mathcal{L}_{MIX}$, inspired by Bandara et al. [7]. For each training batch, this involves shuffling one set of views (e.g., Y_B) along the batch dimension to create $Y_S = \text{Shuffle}(Y_B)$, then generating a mixed view $Y_M = \alpha_{mix} Y_A + (1 - \alpha_{mix}) Y_S$. The mixing coefficient α_{mix} is sampled from a uniform distribution in the unit interval $\alpha_{mix} \sim U(0, 1)$. The embeddings Z_M and Z_S are obtained from their respective views. The mix-up loss penalizes deviations from the assumption that a linear interpolation in the input space corresponds to a linear interpolation in the embedding space.

$$C_{target}^{MA} = \alpha_{mix} (Z_A)^T Z_A + (1 - \alpha_{mix}) (Z_S)^T Z_A$$

$$C_{target}^{MS} = \alpha_{mix} (Z_A)^T Z_S + (1 - \alpha_{mix}) (Z_S)^T Z_S$$

$$\mathcal{L}_{MIX} = \frac{1}{2} (\|C^{MA} - C_{target}^{MA}\|_F^2 + \|C^{MS} - C_{target}^{MS}\|_F^2)$$

where $C^{MA} = (Z_M)^T Z_A$ and $C^{MS} = (Z_M)^T Z_S$ are the actual cross-correlation matrices from the model output. The total loss function optimized during training is a weighted sum:

$$\mathcal{L}_{total} = \mathcal{L}_{BT} + \lambda_{mix} \mathcal{L}_{MIX}$$

where λ_{mix} controls the strength of the mix-up regularization. We found $\lambda_{mix} = 1.0$ (as recommended by [7]) to be effective based on our experiments.

Crucially, we implement a *global shuffling* strategy that randomizes d-pixels across all geographic tiles before batching, smoothing the loss curve, preventing spatial auto-correlation, and improving generalization (see the supplementary materials).

3.4. Inference

Following pretraining, the TESSERA dual encoder (with frozen weights and excluding the projector) is used to generate an annual 128-dimensional embedding for every 10m pixel globally. This involves making inferences from a sample of length 40 from the annual Sentinel-1 and Sentinel-2 time series for each pixel through the trained encoders. To ensure that our maps are gap-free, random sampling with replacement is used to generate augmentations from d-pixels that have fewer than 40 non-cloudy observations. The primary output of this inference stage is a set of annual, global, 10m resolution TESSERA embedding maps. These maps are designed as readily usable multidimensional geospatial data layers. This “Embeddings-as-Data” approach significantly lowers the barrier to entry for end-users, as these rich, precomputed features can be directly ingested by downstream models without the need for raw satellite data processing, running the TESSERA model itself (see supplementary materials). We release embeddings through the pip-installable *GeoTessera* Python library, enabling easy access and integration into downstream workflows.

4. Experiments

4.1. Experimental Setup

Datasets. We evaluate TESSERA on six representative benchmarks that span classification, segmentation, and regression tasks. All datasets use multi-temporal Sentinel-1/2 imagery:

- **TreeSatAI-TS:** Tree species classification in Germany.
- **PASTIS-R:** Parcel segmentation in France.
- **Austrian Crop (Classification):** Crop type classification in Austria [1].
- **Biomasssters:** Above-Ground Biomass (AGB) regression in Finland [65].
- **Borneo Canopy Height:** Canopy height regression in Malaysia [18].
- **Austrian Crop (Segmentation):** Crop semantic segmentation in Austria [1].

To ensure comprehensive evaluation, we select two datasets per task whenever possible: one large-scale (e.g., continental or national coverage) and one fine-grained regional dataset. This design validates both geographic transferability and fine-detail modeling capability. High-

resolution multi-temporal S1+S2 datasets with dense labels are scarce; hence, we additionally contribute two new benchmarks: **Austrian Crop** (parcel-level crop mapping) and **Borneo Canopy Height** (LiDAR-calibrated forest structure in Southeast Asia).

Baselines. We compare against a comprehensive suite of representative remote-sensing foundation models (RSFMs): CROMA [31], DOFA [112], Prithvi [78], RemoteCLIP [54], SatlasNet [8], Scale-MAE [73], Spectral-GPT [36], Galileo [92], Skysense [33], Presto [91], and AlphaEarth [13]. We also include standard UNet [75] and ViT [24] baselines for reference.

Implementation. For embedding-based models (TESSERA, Presto, AlphaEarth), we train a 2-layer MLP for pixel-wise classification, and a lightweight UNet for patch-based classification, segmentation, and regression. Patch-based models are fine-tuned using UPerNet heads. All models are evaluated at 1%, 30% and 100% label ratios to study label efficiency. Additional implementation details are in the supplementary material.

4.2. Classification Results

Patch-based classification: TreeSatAI-TS. Table 1 (a1) reports results on TreeSatAI-TS, a national-scale forest species benchmark. TESSERA achieves a state-of-the-art F1-score of **77.96** with full supervision, significantly outperforming all other models, including the next-best competitor, AlphaEarth (76.90)* and the strongest fine-tuned baseline, the UNet Baseline (73.30). This highlights how our pixel-level temporal modeling effectively captures phenological variations, which improve class separability [39]. Notably, TESSERA’s superiority is even more pronounced in the few-shot setting: at 1% labels, it achieves an F1 score of **60.58**, surpassing the second-best model, AlphaEarth (52.79), by nearly 8 points.

Pixel-wise crop classification: Austrian Crop. On the **Austrian Crop** dataset (Table 1 (a2)), TESSERA demonstrates strong performance. As visualized in Figure 4 (a), it consistently surpasses Presto and AlphaEarth in all standard label ratios (1%-30%). The performance gap is most pronounced in the low-data regime: at just 1% labeled data, TESSERA achieves a **66.15** F1 score. This outperforms AlphaEarth (37.22) by more than 28.9 points and Presto (32.74) by more than 33.4 points. We further investigate extreme label scarcity in Figure 4 (b), evaluating performance with only 1 to 20 labeled samples per class. In this challenging few-shot scenario, TESSERA’s performance remains robust and scales gracefully (e.g., ~ 0.5 F1 with just 4 samples/class), while the F1 scores of other models are below 0.4.

*The TESSERA encoder (45.7M parameters) used during these tasks is also significantly smaller than AlphaEarth’s (480M parameters).

Table 1. **Results across tasks.** (a) **Classification:** F1 $\uparrow$ on TreeSatAI-TS and Austrian Crop (left column; TreeSatAI-TS shows *original/finetuned*). (b) **Segmentation:** mIoU $\uparrow$ on PASTIS-R and Austrian Crop (*original/finetuned* for CROMA-ViT Baseline; single evaluation for Presto, AlphaEarth, TESSERA). (c) **Regression:** RMSE $\downarrow$ on Biomasssters and Borneo CHM (*original/finetuned* for baselines; single evaluation otherwise). **best; second.**

(a1) Classification — TreeSatAI-TS (F1 $\uparrow$)			
Model	TreeSatAI-TS		
	1%	30%	All
CROMA[31]	39.73/43.14	62.44/63.24	70.21/72.09
DOFA[112]	35.04/40.15	59.23/63.46	66.88/67.20
Prithvi[78]	37.21/41.89	60.01/60.66	65.86/67.04
RemoteCLIP[54]	34.94/37.77	57.52/59.70	68.32/68.03
SatlasNet[8]	33.86/37.83	56.80/61.31	66.27/67.85
Scale-MAE[73]	36.20/40.96	59.26/61.29	70.20/73.12
SpectralGPT[36]	41.82/45.66	59.77/60.64	71.04/72.23
Galileo[92]	39.99/42.38	59.42/61.94	69.44/72.17
Skysense[33]	39.72/41.39	61.88/64.80	69.35/72.24
UNet Baseline[75]	39.88/40.43	60.03/62.22	71.39/73.30
ViT Baseline[24]	40.52/41.70	61.79/65.62	72.24/72.60
Presto[91]	46.02	61.45	67.81
AlphaEarth[13]	52.79	71.94	76.90
TESSERA (Ours)	60.58	75.42	77.96

(a2) Classification — Austrian Crop (F1 $\uparrow$)							
Model	Austrian Crop						
	1%	3%	5%	7%	10%	15%	20%
Presto[91]	32.74	37.63	40.26	41.50	43.94	46.01	49.74
AlphaEarth[13]	37.22	40.06	42.73	45.08	48.01	50.47	52.47
TESSERA (Ours)	66.15	71.62	74.32	77.22	78.69	80.12	80.79

(b) Segmentation — PASTIS-R / Austrian Crop (mIoU $\uparrow$)						
Model	PASTIS-R			Austrian Crop		
	1%	30%	All	1%	30%	All
CROMA[31]	7.61/12.33	21.62/23.78	36.24/41.29	5.64/5.19	12.89/15.76	17.10/18.28
DOFA[112]	4.82/5.53	19.28/21.74	25.94/29.55	4.90/4.92	10.05/14.00	17.14/17.39
Prithvi[78]	8.24/11.01	24.86/29.51	34.09/39.52	4.36/3.82	11.12/14.35	16.25/18.60
RemoteCLIP[54]	5.50/7.55	11.72/16.40	20.17/21.03	3.52/3.37	7.76/10.12	11.16/15.15
SatlasNet[8]	9.54/11.58	17.86/21.48	21.02/31.38	4.27/4.05	8.45/9.64	15.61/18.97
Scale-MAE[73]	5.57/7.86	16.34/18.48	23.96/28.51	4.48/4.93	7.42/10.31	13.75/15.22
SpectralGPT[36]	11.25/15.95	26.23/29.33	35.10/40.28	4.85/4.17	13.77/15.20	20.58/22.84
Galileo[92]	9.78/11.83	15.63/17.91	27.92/35.28	9.95/9.03	14.24/15.89	22.80/26.16
Skysense[33]	13.42/14.47	28.54/32.54	32.87/36.41	8.93/9.38	15.29/16.12	26.82/31.08
UNet Baseline[75]	13.52/16.44	25.75/30.08	30.16/37.56	10.64/11.40	14.95/15.10	27.00/30.57
ViT Baseline[24]	12.58/15.72	29.43/34.07	37.37/42.57	10.88/10.03	15.26/16.75	30.72/31.77
Presto[91]	-	-	-	14.25	27.93	34.04
AlphaEarth[13]	27.12	46.42	51.08	14.64	21.75	25.70
TESSERA (Ours)	27.54	46.04	50.68	28.20	48.29	53.12

(c) Regression — Biomasssters / Borneo CHM (RMSE $\downarrow$)					
Model	Biomasssters			Borneo CHM	
	1%	30%	All	1%	30%
CROMA[31]	46.58/44.28	39.24/36.80	37.14/35.07	44.52/41.89	31.25/29.67
DOFA[112]	51.86/47.84	45.72/43.59	43.05/41.10	43.96/40.68	28.97/27.93
Prithvi[78]	43.89/39.37	41.49/38.68	41.12/38.88	43.50/44.27	29.42/27.80
RemoteCLIP[54]	61.70/61.01	52.13/51.32	49.21/47.92	43.24/43.10	32.78/30.78
SatlasNet[8]	49.51/44.64	44.24/43.23	41.93/41.36	45.85/43.01	34.01/32.26
Scale-MAE[73]	56.41/51.51	51.02/49.34	47.05/45.17	47.51/46.72	32.00/31.11
SpectralGPT[36]	64.81/62.28	38.54/36.12	36.87/36.12	43.91/43.65	26.03/25.08
Galileo[92]	52.72/49.15	39.55/36.82	38.52/36.92	44.82/42.78	32.18/31.92
Skysense[33]	47.52/42.28	37.51/37.10	32.52/30.78	42.74/40.96	27.42/26.65
UNet Baseline[75]	44.62/41.70	37.92/36.16	36.41/35.17	41.81/38.61	28.01/26.02
ViT Baseline[24]	48.24/49.42	38.56/38.27	37.98/37.52	42.02/39.95	27.87/26.61
Presto[91]	-	-	-	36.43	20.95
AlphaEarth[13]	35.45	32.33	29.59	36.03	20.62
TESSERA (Ours)	32.50	29.15	27.43	34.82	15.18

Table 2. **Model specification.** Multi-temporal and multi-modal (multiple satellite sources) flags, venue, and parameter size. *Params (M)* indicates encoder frozen / fully fine-tuned.

Model	Multi-temporal	Multi-modal	Publication	Params (M)
CROMA	×	✓	NeurIPS'23	49.1 / 704.9
DOFA	×	✓	arXiv'24	40.6 / 151.7
Prithvi	✓	✓	arXiv'24	40.6 / 126.8
RemoteCLIP	×	✓	TGRS'24	40.6 / 128.1
SatlasNet	✓	✓	ICCV'23	33.3 / 121.2
Scale-MAE	×	×	ICCV'23	49.1 / 352.2
SpectralGPT	✓	×	TPAMI'24	183.5 / 268.9
Galileo	✓	✓	ICML'25	39.4 / 125.9
Skysense	✓	✓	CVPR'24	36.5 / 341.5
UNet	×	×	MICCAI'15	14.8 / 14.8
ViT-B/16	×	×	ICLR'21	125.1 / 125.1
Presto	✓	✓	arXiv'23	30.2 / 30.2
AlphaEarth	✓	✓	arXiv'25	30.1 / 30.1
TESSERA (Ours)	✓	✓	-	30.2 / 30.2

4.3. Segmentation Results

Field parcel segmentation: PASTIS-R. On the large-scale PASTIS-R benchmark (Table 1 (b)), TESSERA achieves competitive performance. With full labels, it obtains **50.68** mIoU, ranking second and just behind AlphaEarth (51.08). However, in the low-label regime (1% labels), TESSERA (**27.54**) surpasses AlphaEarth (27.12) and significantly outperforms all fine-tuned baselines, such as the UNet Baseline (16.44) and Skysense (14.47). Unlike some RSFMs that require compute-heavy decoders (>100M parameters), our compact 30M-parameter UNet

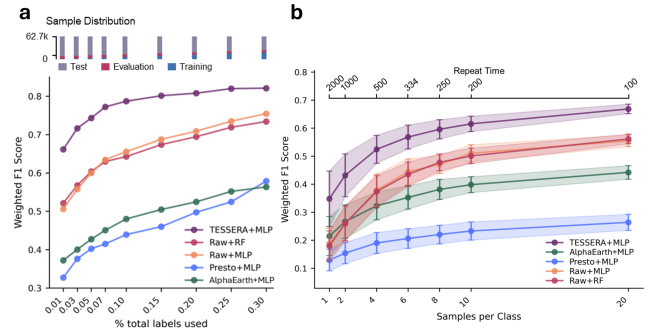


Figure 4. **Label-efficient crop classification on Austrian Crop.** (a) Weighted F1 vs. training label ratio using a small head on frozen embeddings. (b) Few-shot performance. TESSERA attains strong accuracy with very few labels; error bars denote variation over runs.

suffices. This demonstrates that spatial context can be learned efficiently using TESSERA’s pre-trained temporal embeddings.

Regional crop segmentation: Austrian Crop. For the fine-grained Austrian Crop segmentation task, TESSERA achieves **53.12** mIoU, establishing a new SOTA and significantly improving over both AlphaEarth (25.70) by +27.4 points and Presto (34.04) by +19.0 points. This performance gain is consistent across all label regimes, especially

at 1% labels (TESSERA: **28.20** vs AlphaEarth: 14.64), confirming the effective transferability of temporally coherent embeddings. TESSERA yields sharper parcel edges and reduces confusion among visually similar crops (see supplementary materials), indicating its ability to maintain semantic consistency over time.

4.4. Regression Results

Above-ground biomass estimation: Biomasters. As shown in Table 1 (c), TESSERA achieves an RMSE of **27.43** t/ha with full labels – outperforming all other foundation models, including the next-best, AlphaEarth (29.59 t/ha). This result surpasses the strongest fine-tuned RSFM baseline, Skysense (30.78 t/ha). The results demonstrate that temporal embeddings capture biomass dynamics effectively, even under limited supervision. Figure 5 provides a detailed comparison for this task after clipping implausibly high biomass values above 500 t/ha to this threshold. In this analysis, TESSERA (**26.61** t/ha) nearly matches the best-performing supervised task-specific model (25.90 t/ha), which won the Biomasters competition of more than 1000 submissions. The figure further illustrates that TESSERA maintains consistent performance in all label ratios, shows minimal bias in all biomass ranges, and generates spatially coherent AGB maps with only 4% of total labels.

Canopy height regression: Borneo Canopy Height. In the Borneo CHM dataset, TESSERA again demonstrates superior performance, achieving the lowest RMSE of **12.2** m. This result significantly outperforms all other models, including AlphaEarth (16.1 m), Presto (17.9 m), and the best-performing fine-tuned RSFM such as Skysense (15.6 m) and the ViT Baseline (15.8 m). We also compare with DINOv2-derived and Sentinel-2 composite height maps, where TESSERA demonstrates the closest agreement with LiDAR ground truth (see supplementary materials).

5. Discussion & Ablation Studies

We validate our design choices on the Austrian Crop classification task. Table 3 summarizes the results; in the following we organize the findings with brief Q&A headings and defer all methodological details to the supplementary material.

Do shuffling and mixup help? Yes. Removing global shuffling or mixup drops validation F1 by $-9.2/-11.1$ and RankMe score by $-0.116/-0.106$. Removing both leads to the largest fall (-14.7 F1; see Table 3). The two strategies are complementary: shuffling smoothes the loss curve, mitigates spatial autocorrelation while mixup helps to prevent feature overfitting [7].

How robust are the embeddings to cloudiness? We find that when the annual number of non-cloudy *valid* Sentinel-

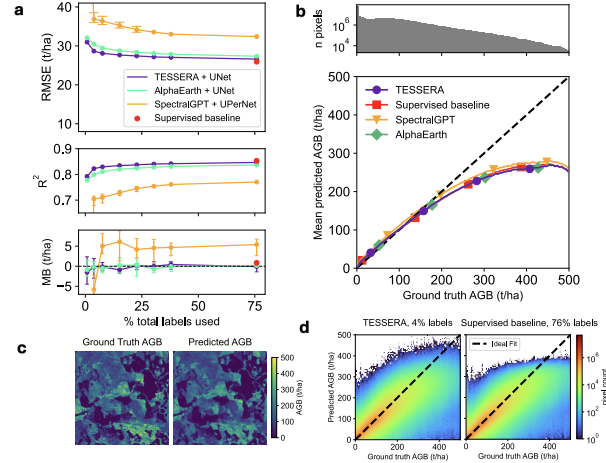


Figure 5. **Biomasters AGB regression.** (a) RMSE/ R^2 /MB (mean bias) vs. label fraction. (b) Predicted AGB vs. Ground truth AGB. (c) Spatial AGB maps from TESSERA trained with 4% labels. (d) Predicted vs. ground-truth scatter. TESSERA closely tracks the task-specific winner with significantly fewer labels.

Table 3. **Ablation on the Austrian Crop dataset.** We report **Validation F1** and **RankMe** (higher $\uparrow$ is better). Parentheses show change vs. the Baseline.

Configuration	Val. F1 ($\uparrow$)	RankMe ($\uparrow$)
Baseline (Full Model)	77.3	0.963
w/o Global Data Shuffling	68.1 (-9.2)	0.847 (-0.116)
w/o Mixup Regularization	66.2 (-11.1)	0.857 (-0.106)
w/o Sentinel-1 Data	74.2 (-3.1)	0.931 (-0.032)
w/o Shuffling & Mixup	62.6 (-14.7)	0.867 (-0.096)
w/o Embedding Quantization	77.9 (+0.6)	0.972 (+0.009)
w/o Pretraining	43.8(-33.5)	-

2 observations drops ≤ 10 , Macro-F1 performance drops sharply (see Figure 6). At a global scale, we find that embeddings are quite robust to cloudiness since there are more than 70 observations per year for each orbit.

Is fine-tuning weights necessary? Self-supervised pre-training of the encoder yields large performance gains compared to randomly initialized weights, a +33.5 F1 improvement in the Austrian Crop classification task. Pretraining, somewhat surprisingly, makes subsequent finetuning unnecessary: Figure 6 (b) shows that finetuning the encoder provides no benefit. This suggests that the embeddings encapsulate a highly generalized representation.

Are regional embeddings necessary? We find that re-training with region-specific data offers negligible gains (Figure 6 c). A single global model is sufficient for high performance across diverse geographies, reinforcing the *Embeddings-as-Data* paradigm and making costly regional retraining unnecessary.

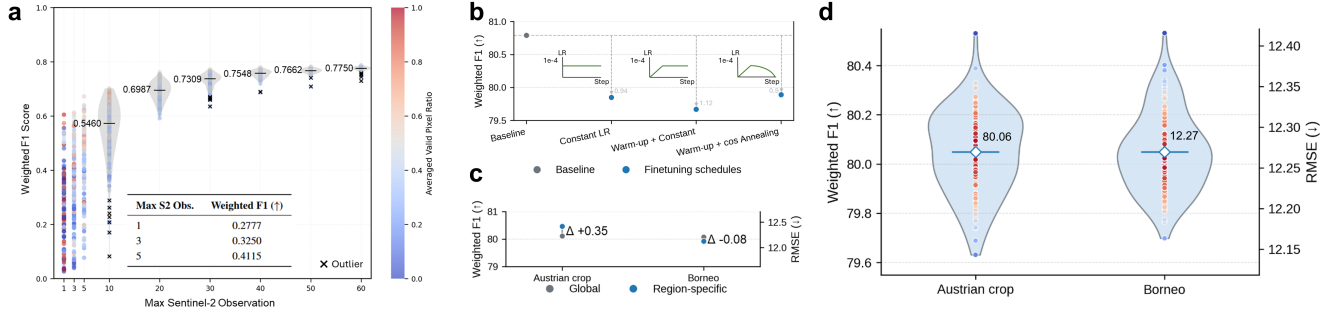


Figure 6. **Robustness and sensitivity analyses.** (a) *Cloud robustness*: performance remains stable until Sentinel-2 valid observations per year drop below ~ 20 , after which accuracy degrades markedly. (b) *Finetuning impact*: evaluating the effect of three different learning rate strategies during finetuning. (c) *Embedding specificity*: comparing the performance of a global embedding versus an embedding retrained on region-specific data. (d) *Temporal sampling uncertainty*: selecting $L=40$ timesteps within a year produces consistently similar results across random choices.

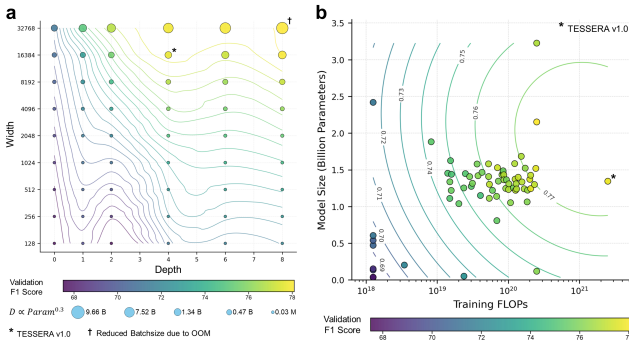


Figure 7. **Scaling behavior.** (a) Projector depth/width sweep: validation F1 improves with capacity; † marks the max-F1 setup under memory constraints; * denotes the chosen **TESSERA v1.0**. (b) Validation F1 scales predictably with training FLOPs and parameters, with **v1.0** near the compute-accuracy frontier.

Can a spatial head recover spatial context? Yes. Although pretraining is pixel-wise (with no explicit spatial structure), adding a spatial head (e.g., a UNet) consistently improves segmentation and regression tasks, with larger gains at higher label budgets. This suggests that spatial cues can be injected *downstream* efficiently without complicating pretraining or creating purely pixel-based embeddings.

How sensitive are embeddings to random temporal sampling? Not much. With $L=40$ temporal samples, repeated random draws produce tightly clustered results (low variance), consistent with our time-sampling invariance objective (shown in Figure 6 c). We also compare $L \in \{20, 40, 96, 365\}$ and find that $L=40$ is the accuracy–efficiency sweet spot (see supplementary material).

Scaling laws. Figure 7 shows that the performance scales predictably with the depth/width of the projector, FLOPs, and the parameters; **TESSERA v1.0** (star) lies near the optimal compute–accuracy frontier, while the dagger (†) marks the maximum-F1 configuration constrained by memory/batch size.

Flexibility of the inference time window. In the default setting, **TESSERA** is trained using a one-year temporal window. However, this one-year window is not a strict requirement. We find that applying the model to shorter seasonal windows during inference (e.g., only from spring to autumn) still yields effective embeddings (see supplementary materials for seasonal window analyses on the Austrian crop dataset).

Does `int8` quantization significantly degrade performance? No: there is only a small drop in performance. Removing quantization slightly *improves* metrics (e.g., $+0.6$ F1 and $+0.009$ RankMe in Table 3), indicating that `int8` introduces minor information loss. However, using `int8` compresses the embeddings to $\sim 25\%$ of the `fp32` storage footprint (and reduces I/O), which is attractive for global-scale products and for use on an edge device.

6. Conclusion

We introduced **TESSERA**, a pixel-wise foundation model for Earth Observation that learns robust representations from sparse, multi-modal time series via temporal sampling invariance. Extensive experiments show that **TESSERA** establishes a new state-of-the-art across diverse tasks with unparalleled label efficiency. We pioneer the “Embeddings-as-Data” paradigm, releasing analysis-ready global embeddings via the **GeoTessera** library to lower barriers for geospatial analysis and empower future research.

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